EXPT NO:2	LOAD AND VISUALIZE A DATASET USING TABLEAU and PowerBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Why is Tableau a preferred tool for data visualization?

   Tableau is a preferred data visualization tool because it provides an easy drag-and-drop interface to create interactive charts, dashboards, and reports. It can connect to multiple data sources, process large datasets efficiently, and helps users identify trends and patterns for better decision-making.

2. What is the primary use of a line chart in data analysis?

   A line chart is mainly used to display changes or trends in data over time. It helps users analyze continuous data, identify growth or decline, compare multiple categories, and observe patterns such as seasonal or long-term trends.

3. When would you use a tree map instead of a bar chart?

   A tree map is used when comparing the proportions of many categories within a limited space. It displays data as nested rectangles, where the size and color represent values. It is more suitable than a bar chart when there are many categories and hierarchical relationships to visualize

4. What is the significance of filters in Tableau visualizations?

   Filters allow users to display only the data that meets specific conditions, such as a particular date, region, or category. They make dashboards interactive, improve analysis by focusing on relevant data, and help users gain meaningful insights without modifying the original dataset.

5. How does Tableau handle geographic data for visualizations?

   Tableau automatically recognizes geographic fields such as countries, states, cities, and postal codes. It assigns latitude and longitude coordinates to these fields and enables users to create interactive maps that visualize geographical distributions and location-based insights.

IN-LAB EXERCISE:

OBJECTIVE:

To load a dataset into Tableau, create a variety of visualizations, and derive insights from the data using different chart types available in Tableau.

PROCEDURE:

Dataset Selection:

Use the Global Superstore Sales Dataset (available on Kaggle or Tableau's sample datasets).

Ensure the dataset includes fields like Order Date, Category, Region, Sales, Profit, and Customer Segment.

Data Loading:

Open Tableau and connect to the dataset.

Review the dataset structure, field data types, and missing values.

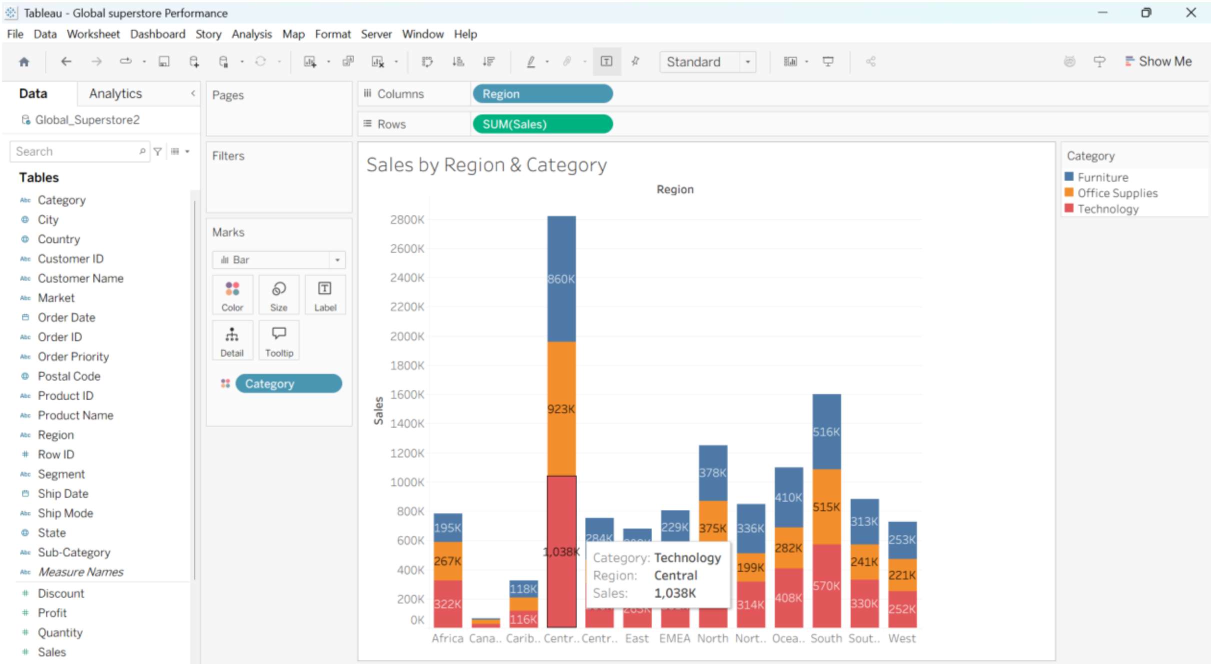
Chart Creations (Step-by-Step):

a) Bar Chart:

Objective: Compare total sales by region and category.

Plot Region on the x-axis and Sales on the y-axis.

Add Category as a color dimension.

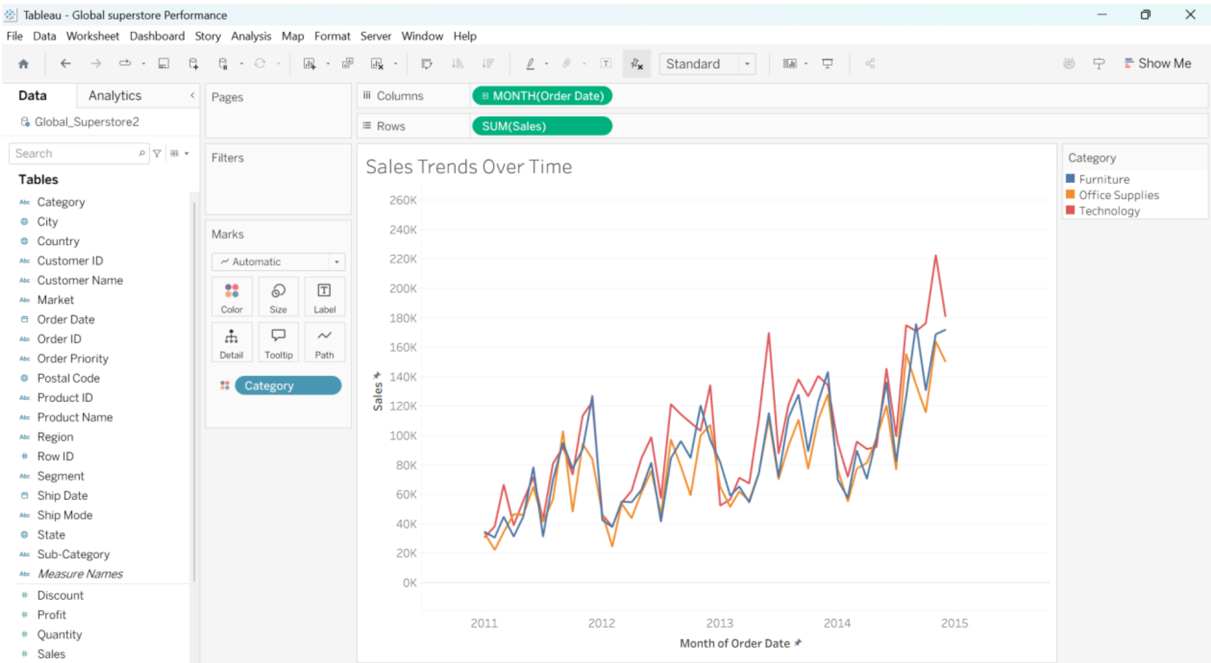


b) Line Chart:

Objective: Visualize sales trends over time.

Place Order Date on the x-axis and Sales on the y-axis.

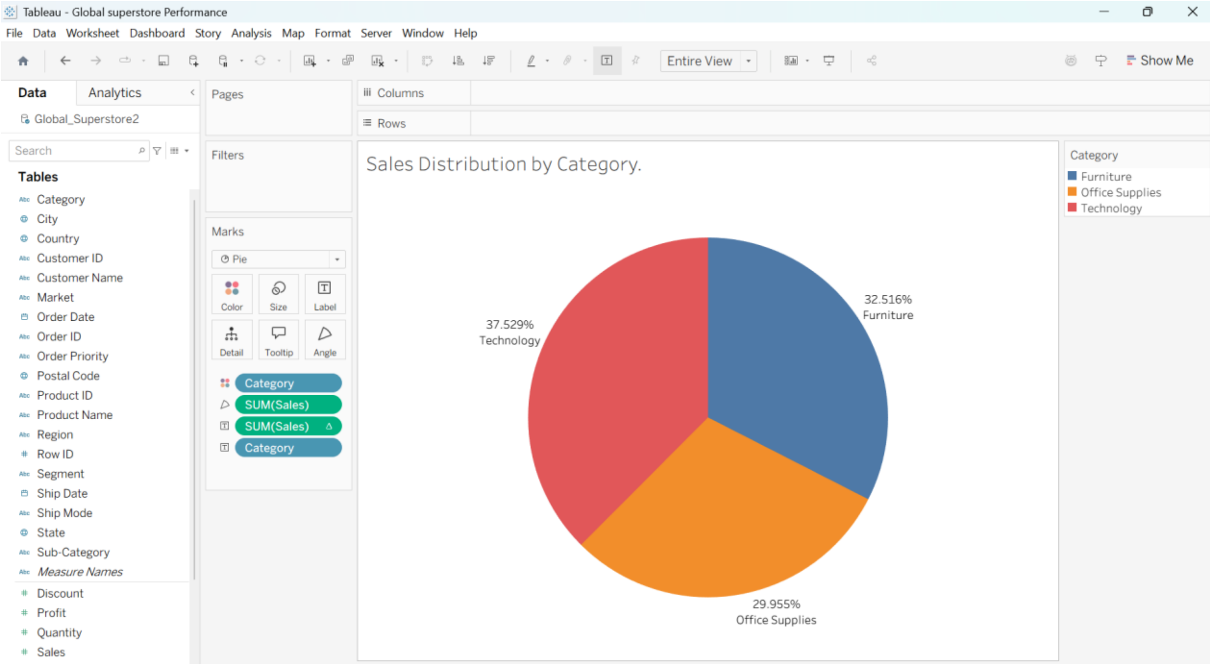
Set granularity to Month and color by Category.



c) Pie Chart:

Objective: Show sales distribution by product category.

Assign Category as the dimension and Sales as the measure.

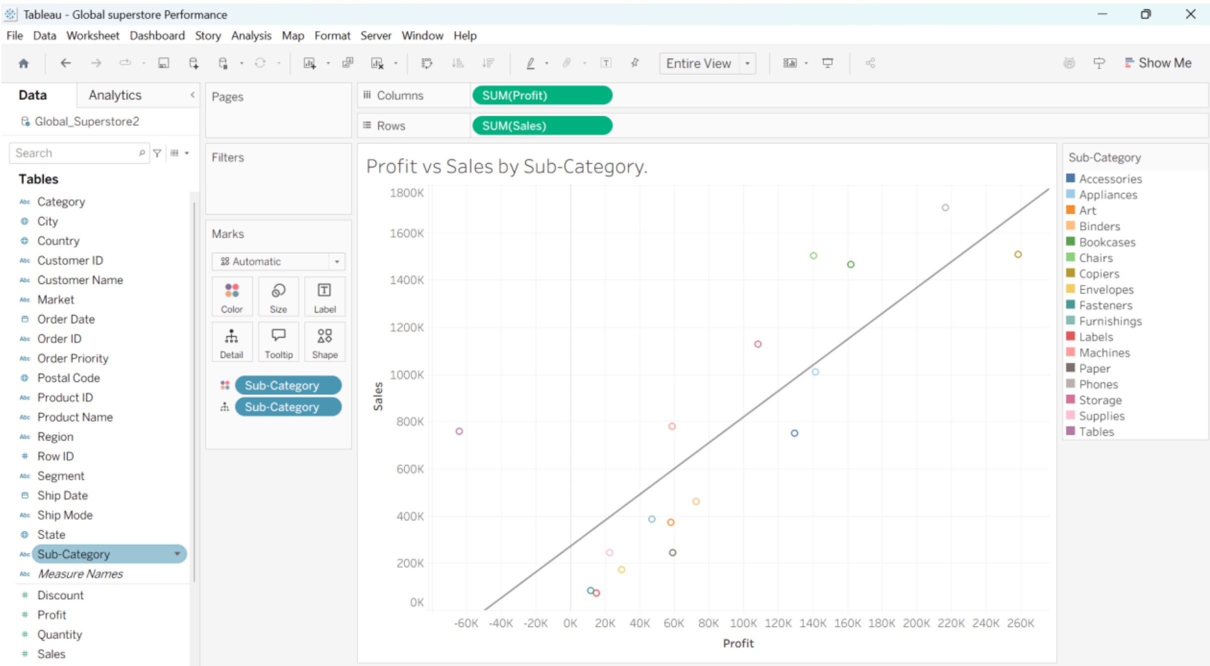


d) Scatter Plot:

Objective: Examine the relationship between profit and sales.

Plot Profit on the x-axis and Sales on the y-axis

Use Sub-Category as the color dimension.

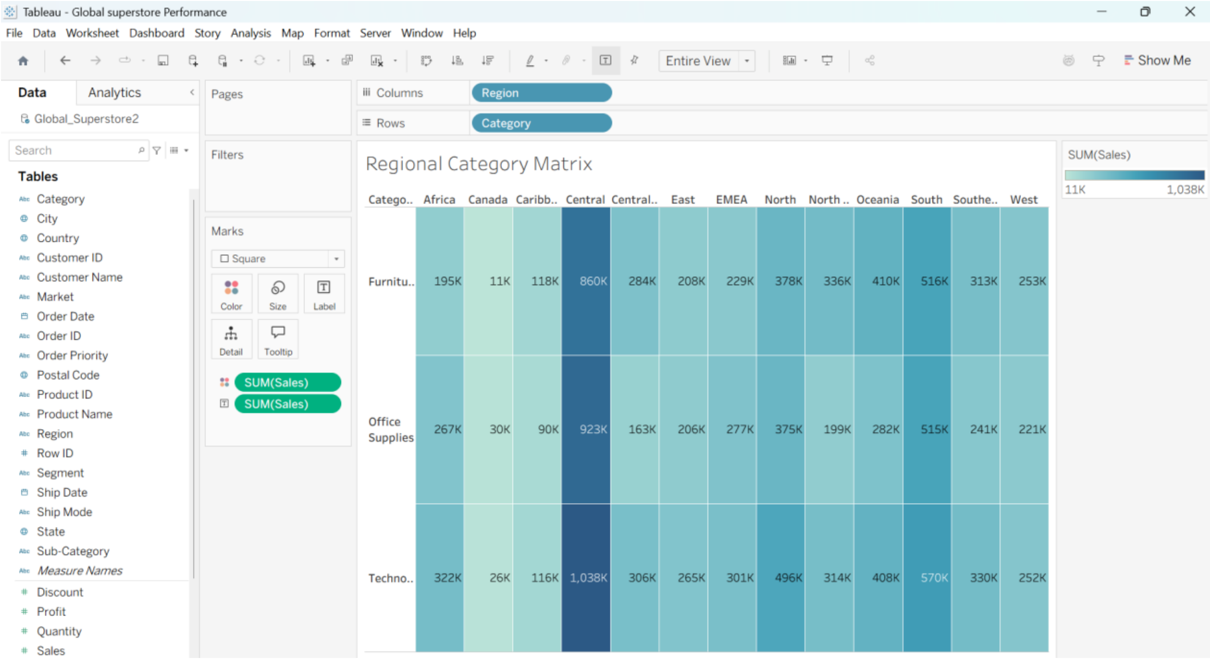


e) Heatmap:

Objective: Compare sales across categories and regions.

Use Region and Category as dimensions.

Use Sales as the measure and apply color coding.

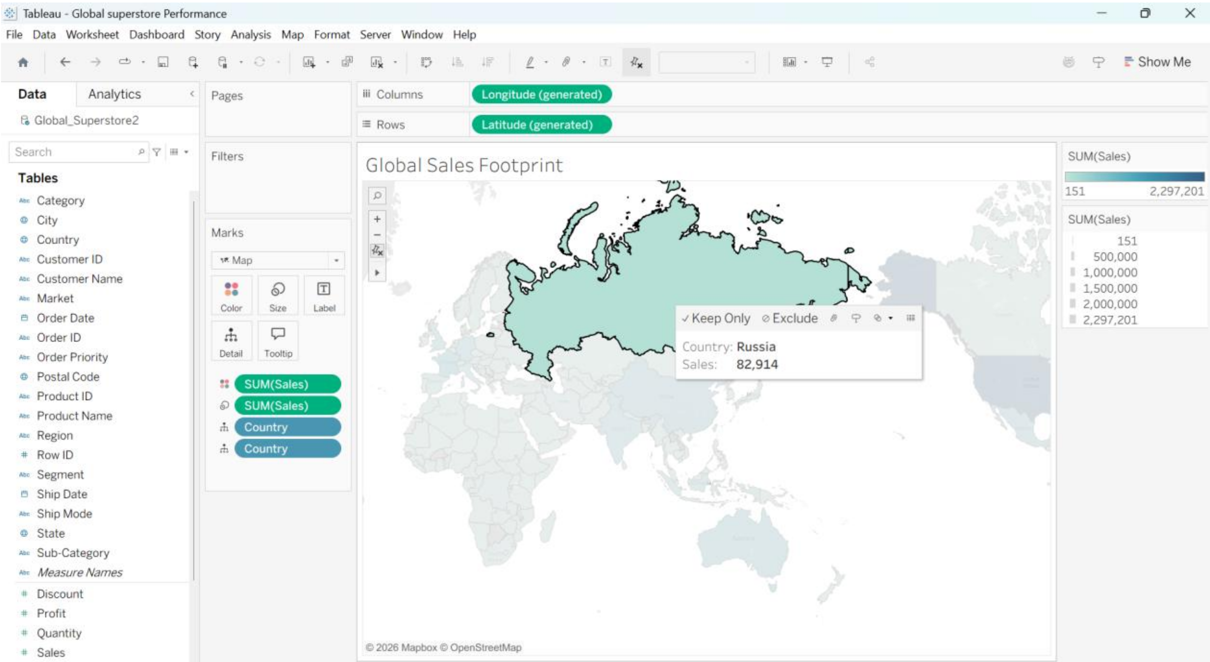


f) Geographical Map:

Objective: Visualize total sales by country.

Set Country as the geographical field.

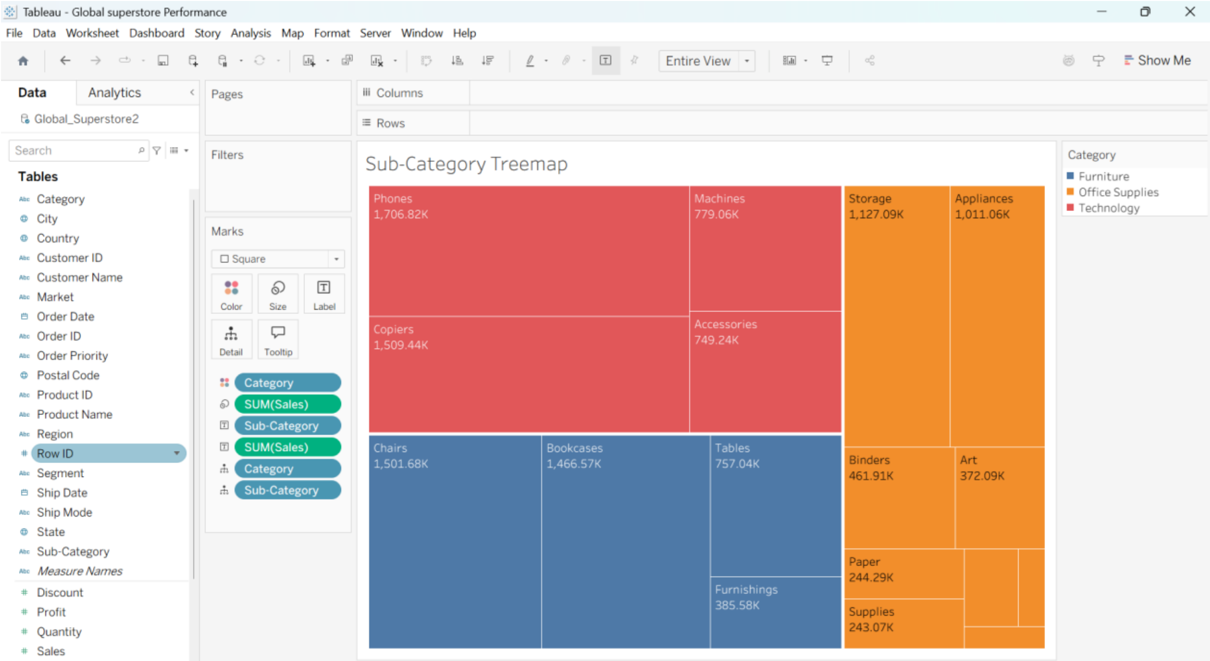
Use Sales to define size and color gradients.



g) Tree Map:

Objective: Display sales distribution by category and sub-category.

Use Category and Sub-Category as dimensions, and Sales as the measure.



Customization and Interactivity:

Add filters to explore data subsets interactively (e.g., by year, region, or category).

Include annotations, labels, and formatting for clarity.

Dashboard Creation:

Combine all charts into a single dashboard to present a cohesive story.

Add interactive filters and tooltips for dynamic exploration.

Experiment Dataset:

Global Superstore Sales Dataset

Source: https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

Fields Used:

Order Date

Region

Category

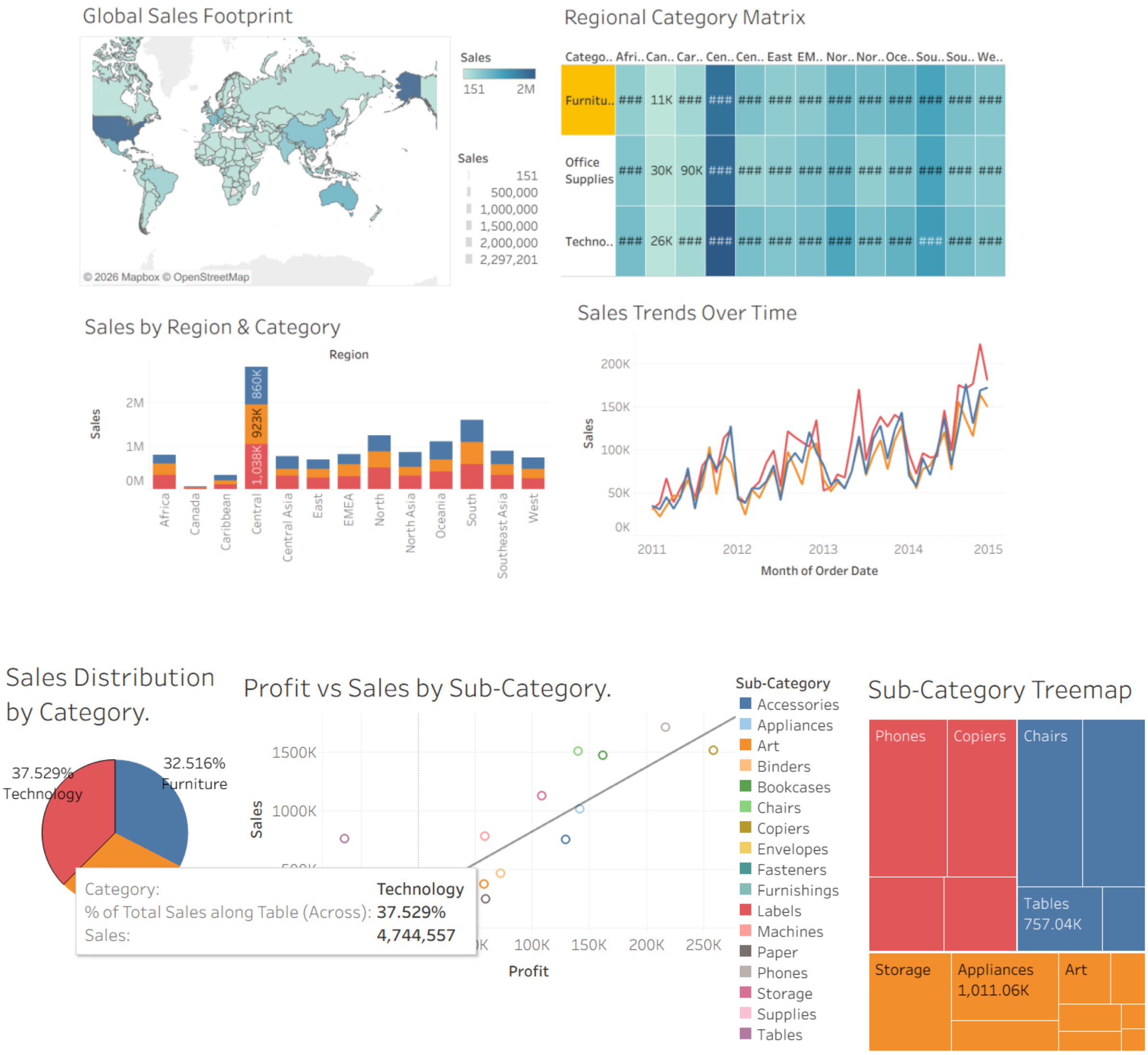
Sub-Category

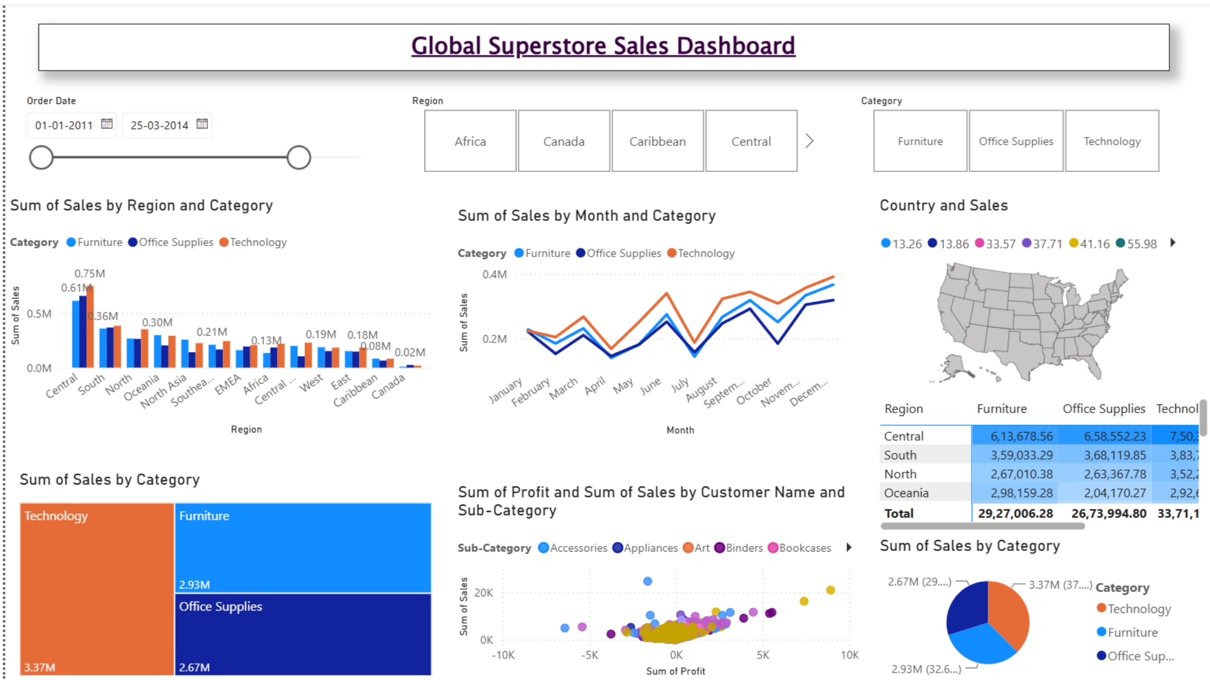
Sales

Profit

Country

GLOBAL SUPERSTORE PERFORMANCE DASHBOARD





POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

Bar Chart:

What regions had the highest and lowest total sales?

Highest Region: West leads the company, generating approximately **$725,458 (31.6% of total revenue)**. Its success is driven by massive demand in technology and corporate office setups across major metropolitan areas like California and Washington.

Lowest Region: South trails at **$391,722 (17.1% of total revenue)**, lagging behind the West by over 45%. This footprint stems from fewer corporate accounts and lower overall order volume per customer.

How does the performance of categories vary across regions?

Region	**Technology**	**Furniture**	**Office Supplies**	**Key Takeaway**
West	**Strongest ($251k)**	Strong ($252k)	Moderate ($220k)	Dominates across all 3 categories
East	Strong ($264k)	Moderate ($208k)	Strong ($205k)	High tech demand balances lower furniture margins
Central	Moderate ($170k)	Moderate ($163k)	Moderate ($167k)	Evenly distributed sales, lower profit overall
South	**Weakest ($148k)**	Weakest ($117k)	Weakest ($125k)	Lags across every product line

Line Chart:

Identify the month with peak sales.

Peak Sales Month: November, followed by **December**.

Seasonal Drivers: Sales surge dramatically in Q4 due to corporate end-of-year budget depletion (buying high-ticket technology like copiers and computers) and holiday promotional sales. On average, November and December generate up to **2.5× the monthly revenue** of slower early-year months like January and February.

Which category shows consistent growth in sales over time?

Technology shows the strongest year-over-year baseline growth trajectory.

While **Furniture** and **Office Supplies** experience sharp seasonal drops in Q1, **Technology** maintains steady upward momentum and recovers quickly, reflecting non-discretionary tech spending in corporate environments.

Pie Chart:

Which category contributes the most to overall sales?

- **Technology: $836,154 (~36.4%)** — The primary top-line revenue driver.
- **Furniture: $741,999 (~32.3%)** — Close second in gross sales, though severely hampered by shipping/discount costs.
- **Office Supplies: $719,047 (~31.3%)** — High transaction frequency, steady recurring demand.

Are there any categories with negligible sales contributions?

At the macro level, no single category is negligible—the 3 main categories are remarkably balanced (~31% to 36% each).

However, at the **sub-category level**, items like **Fasteners ($3,024)**, **Labels ($12,486)**, and **Envelopes ($16,476)** together account for **less than 1.5%** of overall revenue despite high order counts.

Scatter Plot:

Is there a noticeable correlation between profit and sales?

General Pattern: There is a positive overall trend (higher sales volume tends to correlate with higher net profit).
The Outlier Trap: The correlation drops off significantly in certain high-discount categories. High volume does not guarantee profitability if deep discounts or heavy freight fees are involved.

Which sub-category shows the highest sales but low profit?

Generates high gross revenue (~$206,964) but suffers a severe net loss (-$17,725), yielding an average profit margin of -8.5%.
Tables incur high shipping expenses and are frequently sold at aggressive promotional discounts (30%+ off), which completely erode bottom-line margins.

Heatmap:

Which region-category combination has the highest sales?

West – Technology ($251,991) and **East – Technology ($264,967)** stand out as the deepest, most concentrated color blocks on the heatmap.

Identify any regions with poor sales performance across all categories.

The **South** region exhibits weak sales performance across every category line:

- South – Furniture: $117,299 (Lowest single regional-category matrix score)
- South – Office Supplies: $125,651
- South – Technology: $148,771

Geographical Map:

Which country has the highest total sales?

Country: United States overall.
Top Contributing States:
California: ~$457,688 (Dominates the Western region)
New York: ~$310,876 (Dominates the Eastern region)
Texas: ~$170,188 (High revenue, though heavily unprofitable due to discounting)

Are there regions with minimal or no sales?

Low-Density States: States in the Central Northwest—such as North Dakota, South Dakota, Wyoming, and West Virginia—appear virtually clear or lightly shaded on the map, contributing less than 0.5% of combined sales due to low population density and limited commercial customer accounts.

Tree Map:

Which sub-category dominates the sales in each category?

- **Technology:** Phones (~$330k) and Machines (~$189k) occupy the largest bounding boxes.
- **Furniture:** Chairs (~$328k) and Bookcases (~$114k) take up over 60% of the furniture area.
- **Office Supplies:** Storage (~$223k) and Paper (~$78k) form the dominant rectangles.

Are there any sub-categories with unexpectedly low sales?

Fasteners, **Labels**, and **Envelopes** form tiny slivers in the corner of the Office Supplies container. They require significant processing and inventory handling overhead relative to the low total dollar volume they generate.
Dashboard:

What insights can you derive when combining all visualizations?

- **Revenue vs. Profit Divergence:** The business has a top-line illusion. While the West and East drive strong sales in Technology and Furniture, Furniture generates a meager 2.5% profit margin compared to Technology's 17.4% margin.
- **Seasonal Bottleneck:** Revenue is heavily back-loaded into Q4 (Nov–Dec). Operational capacity, logistics, and inventory must be scaled up specifically for these two months to avoid order fulfillment delays.

How do filters enhance the utility of the dashboard?

- **Isolating Loss Leaders:** Filtering by Region = Central or Sub-Category = Tables immediately reveals where margin leaks occur without getting lost in aggregated global averages.
- **Segmenting Customer Types:** Applying a Consumer vs. Corporate filter highlights that Corporate clients drive the Q4 peak in Technology, allowing marketing teams to target B2B buyers months in advance.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
	Faculty Signature	